

Weekly Putnam

3 October 2026

PROBLEM

Determine all positive integers n for which there exist $n \times n$ matrices A_1, A_2, A_3, A_4, A_5 with real entries such that

$$A_i^2 = 0 \text{ for every } i \quad \text{and} \quad A_i A_j + A_j A_i = I \text{ for all } i \neq j.$$

Here I is the $n \times n$ identity matrix.

Answer: all multiples of 8 · Level: A5 · Topic: Linear algebra

Hints

1. For $x \in \mathbb{R}^5$ put $A(x) = \sum x_i A_i$. Show that $A(x)^2$ is always a scalar multiple of I , and identify the quadratic form in x that appears.
2. That form has one positive and four negative directions. Use it to find real combinations D, C, F, G of the A_i that anticommute pairwise, with $D^2 = I$ and $C^2 = F^2 = G^2 = -I$. Two anticommuting square roots of $-I$ already force $4 \mid n$.
3. The product CFG squares to I , commutes with C and F , and anticommutes with D . Compare the two eigenspaces of CFG .

The key idea

Linear combinations of the five square-zero matrices obey the same rules as the quadratic form $q(x) = \sum_{i < j} x_i x_j$, which has signature $(1, 4)$. Diagonalising q turns the problem into one about anticommuting matrices squaring to $\pm I$. Two of them make each invariant space a quaternionic space, which gives a factor 4; the involution CFG splits $\mathbb{R}^n$ into two halves of equal size, each quaternionic, which gives the last factor 2. The fifth matrix costs nothing extra: an explicit 8×8 example carries all five.

Solution

Answer. Exactly the multiples of 8.

Step 1: the quadratic form. For $x \in \mathbb{R}^5$ write $A(x) = \sum_{i=1}^5 x_i A_i$ and $s(x) = x_1 + \cdots + x_5$. Expanding and using the hypotheses,

$$A(x)A(y) + A(y)A(x) = \sum_{i \neq j} x_i y_j I = (s(x)s(y) - x \cdot y)I. \quad (1)$$

In particular $A(x)^2 = q(x)I$ with $q(x) = \frac{1}{2}(s(x)^2 - |x|^2)$, and $A(x)$ anticommutes with $A(y)$ exactly when $s(x)s(y) = x \cdot y$.

Step 2: anticommuting generators. Define

$$D = A_1 + A_2, \quad C = A_1 - A_2, \quad F = A_3 - A_1 - A_2, \quad G = \frac{1}{\sqrt{3}}(2A_4 - A_1 - A_2 - A_3).$$

Their coefficient vectors are $(1, 1, 0, 0, 0)$, $(1, -1, 0, 0, 0)$, $(-1, -1, 1, 0, 0)$ and $\frac{1}{\sqrt{3}}(-1, -1, -1, 2, 0)$,

with sums 2, 0, -1 , $-1/\sqrt{3}$. Formula (1) gives at once

$$D^2 = I, \quad C^2 = F^2 = G^2 = -I,$$

and for each of the six pairs one checks $s(x)s(y) = x \cdot y$ (for example, for D, F both sides equal -2 ; for F, G both equal $1/\sqrt{3}$). So D, C, F, G anticommute pairwise.

Lemma (quaternionic spaces). *Let $W \neq 0$ be a finite-dimensional real vector space and $P, Q: W \rightarrow W$ linear maps with $P^2 = Q^2 = -1$ and $PQ = -QP$. Then $4 \mid \dim W$.*

Proof. Put $R = PQ$. Then $R^2 = -1$ and P, Q, R anticommute pairwise, so for real a, b, c, d

$$(a + bP + cQ + dR)(a - bP - cQ - dR) = (a^2 + b^2 + c^2 + d^2) \cdot 1.$$

Hence for $w \neq 0$ the four vectors w, Pw, Qw, Rw are linearly independent, and

$$Hw := \text{span}\{w, Pw, Qw, Rw\}$$

is a 4-dimensional subspace invariant under P and Q . Any nonzero $u \in Hw$ satisfies $Hu \subseteq Hw$ and $\dim Hu = 4$, so $Hu = Hw$: the only invariant subspaces of Hw are 0 and Hw . Now let U be an invariant subspace and $w \notin U$. Then $Hw \cap U$ is invariant and does not contain w , so it is 0, and $U \oplus Hw$ is invariant of dimension $\dim U + 4$. Starting from $U = 0$ we reach $U = W$, so $\dim W$ is a multiple of 4. $\square$

Step 3: necessity. Let $\omega = CFG$. Each of C, F, G anticommutes with the other two, so moving it across the product CFG produces two sign changes: ω commutes with C, F and G . The matrix D anticommutes with all three factors, so $D\omega = -\omega D$. Moreover, since FG commutes with C ,

$$\omega^2 = C(FG)C(FG) = C^2(FG)^2 = (-I)(-F^2G^2) = (-I)(-I) = I.$$

Thus $\mathbb{R}^n = V_+ \oplus V_-$ with $V_{\pm} = \ker(\omega \mp I)$. If $\omega v = v$ then $\omega(Dv) = -D\omega v = -Dv$, so D maps V_+ into V_- and V_- into V_+ ; as D is invertible ($D^2 = I$), $\dim V_+ = \dim V_- = n/2$. In particular n is even and $V_+ \neq 0$. Since C and F commute with ω , they map V_+ to itself, and on V_+ they satisfy the hypotheses of the Lemma. Hence $4 \mid n/2$, that is, $8 \mid n$. (Only four of the five matrices were used.)

Step 4: an example for $n = 8$. Let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

so that $X^2 = Z^2 = I$, $J^2 = -I$, and any two of X, Z, J anticommute. For Kronecker products $P_1 \otimes P_2 \otimes P_3$ of I, X, Z, J , two such products anticommute exactly when the number of positions holding two different non-identity factors is odd. Define the 8×8 matrices

$$D_0 = I \otimes I \otimes X, \quad F_1 = I \otimes I \otimes J, \quad F_2 = I \otimes J \otimes Z, \quad F_3 = J \otimes X \otimes Z, \quad F_4 = J \otimes Z \otimes Z.$$

Then $D_0^2 = I$, $F_k^2 = -I$ (each F_k has exactly one factor J), and the five matrices anticommute pairwise (each of the ten pairs differs in exactly one position with two different non-identity factors). Consequently, for $P = aD_0 + \sum b_k F_k$ and $P' = a'D_0 + \sum b'_k F_k$,

$$PP' + P'P = 2(aa' - b \cdot b')I. \tag{2}$$

Now set

$$A_k = \frac{1}{\sqrt{2}}(D_0 + F_k) \quad (k = 1, 2, 3, 4), \quad A_5 = \frac{1}{\sqrt{2}}(2D_0 + F_1 + F_2 + F_3 + F_4).$$

By (2): $A_k^2 = \frac{1}{2}(1 - 1)I = 0$ and $A_5^2 = \frac{1}{2}(4 - 4)I = 0$; for $k \neq l \leq 4$, $A_k A_l + A_l A_k = 2 \cdot \frac{1}{2}(1 - 0)I = I$; and $A_k A_5 + A_5 A_k = 2 \cdot \frac{1}{2}(2 - 1)I = I$. So $n = 8$ works.

Step 5: all multiples of 8. If $n = 8m$, take the block-diagonal matrices $\text{diag}(A_i, \dots, A_i)$ with m blocks. All relations hold blockwise.

Hence the required n are exactly $8, 16, 24, \dots$

Why it is hard, and why it is original

Why the first ideas fail. The two classical moves each give too little. Working with two of the matrices only yields $2 \mid n$: this is the folklore fact that $A^2 = B^2 = 0$ and $AB + BA = I$ force even size, since then $\mathbb{R}^n = \ker A \oplus \ker B$ and A maps $\ker B$ isomorphically onto $\ker A$. Counting independent products is the other natural attempt: the sixteen products of subsets of $\{D, C, F, G\}$ are linearly independent (their traces vanish), so $n^2 \geq 16$ and $n \geq 4$. That bound is not sharp. The real obstruction is arithmetic over $\mathbb{R}$: the four generators carry one quaternionic structure and one extra involution, and that is what produces 8.

The trap in the count. The minimal sizes for 2, 3, 4 such matrices are 2, 4, 8, so the doubling pattern suggests 16 for five. It is 8: a fifth matrix fits into the same 8-dimensional space. Over $\mathbb{C}$ the answer would be different (four matrices fit in size 4), so the reality of the entries is essential, and the solver has to find an explicit real example rather than quote a complex one.

Originality and nearest relatives. The nearest known statement is the textbook exercise that $A^2 = B^2 = 0$, $AB + BA = I$ forces n even, which is the two-matrix case and needs only Step 3's swapping idea. Families of matrices that anticommute and square to $\pm I$ are classical (they lie behind the Hurwitz–Radon theory of sums of squares). This problem differs in its form: the given matrices are square-zero and all their anticommutators equal I , so the anticommuting structure is hidden until the quadratic form of Step 1 is diagonalised, and the answer asks for an exact arithmetic classification rather than a bound.

Verification

The construction of Step 4 was checked in exact integer arithmetic: with $B_i = \sqrt{2} A_i$ (integer matrices), $B_i^2 = 0$ and $B_i B_j + B_j B_i = 2I$ for all ten pairs. The generators D, C, F, G of Step 2 were built from these A_i and checked to satisfy $D^2 = I$, $C^2 = F^2 = G^2 = -I$, pairwise anticommutation, $\omega^2 = I$ and $D\omega = -\omega D$, with $\text{tr } \omega = 0$ as the halving predicts. A general real Clifford-type construction in sizes 2, 32 and 512 (for 2, 10 and 18 matrices) was also checked, together with the full rank of all 1024 products in size 32. As an independent check that size 4 fails for five matrices, 40 random-start nonlinear least-squares searches over 4×4 real matrices all stalled at a residual bounded well away from zero (best value 1.2), while the explicit size-8 example is exact.